THE CHINESE UNIVERSITY OF HONG KONG, SHENZHEN
School of Management and Economics

ECONOMICS 3121
Introductory Econometrics

Winter term 2019 **MIDTERM EXAM** Yi Ding

DATE: **Saturday March 23, 2019**

TIME: **150 minutes; 2:00 p.m. – 4:30 p.m.**

INSTRUCTIONS: The exam consists of **FIVE (5)** questions. Students are required to answer **ALL FIVE (5)** questions.

Answer all questions in the exam booklets provided. Be sure your ***name*** and ***student number*** are printed clearly on the front of all exam booklets used.

Do not write answers to questions on the front page of the first exam booklet.

Please label clearly each of your answers in the exam booklets with the appropriate number and letter.

Please write legibly.

This exam is **CONFIDENTIAL**. This question paper must be submitted in its entirety with your answer booklet(s); otherwise your exam will not be marked.

A formula sheet and a table of percentage points of the t-distribution is given on the last pages of the exam.

MARKING: The marks for each question are indicated in parentheses immediately above each question. **Total marks** for the exam **equal 130**.

GOOD LUCK!

Question 1: Multiple Choice (44 points, 2 points each)
Select the **BEST** response to each of the following questions.

1. Which of the following correctly identifies a difference between cross-sectional data and time series data?
   a) Cross-sectional data is based on temporal ordering, whereas time series data is not.
   b) Time series data is based on temporal ordering, whereas cross-sectional data is not.
   c) Cross-sectional data consists of only qualitative variables, whereas time series data consists of only quantitative variables.
   d) Time series data consists of only qualitative variables, whereas cross-sectional data does not include qualitative variables.

Answer: b

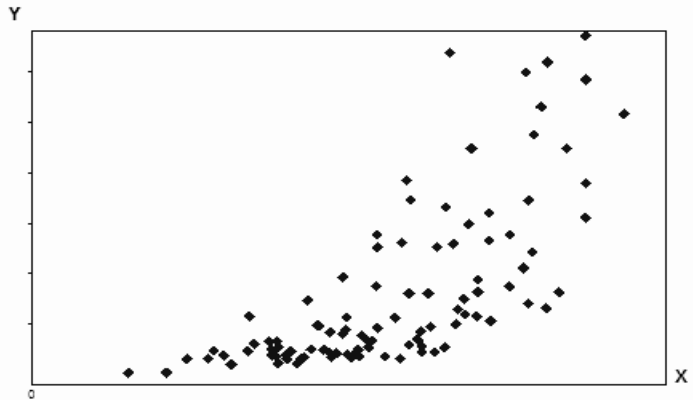
2. Which of the following is NOT correct with regard to the *p*-value attached to a test statistic?
   a) *p*-values can only be used for two-sided tests
   b) It is the marginal significance level where we would be indifferent between rejecting and not rejecting the null hypothesis
   c) It is the exact significance level for the test
   d) Given the *p*-value, we can make inferences without referring to statistical tables

Answer: a

3. The slope estimator, $\hat{\beta}_1$, has a larger standard error, other things equal, if
   a) there is more variation in explanatory variable, x.
   b) there is a smaller variance of error term, u.
   c) the sample size is smaller.
   d) the intercept estimator, $\hat{\beta}_0$, is larger.

Answer: c

4. Which regression model would best fit the data shown in the plot below?
   a) $y = \beta_0 + \beta_1 x + u$
   b) $y = \beta_0 + \beta_1 \log(x) + u$
   c) $\log(y) = \beta_0 + \beta_1 x + u$
   d) $\log(y) = \beta_0 + \beta_1 \log(x) + u$



Answer: c

5. Which of the following statements is true when the dependent variable, $y > 0$
   a) Taking log of a variable often expands its range.
   b) Models using $log(y)$ as the dependent variable may satisfy CLM assumptions more closely than models using the level of y.

c) Taking log of variables make OLS estimates more sensitive to extreme values.
d) Taking logarithmic form of variables make the slope coefficients more responsive to rescaling.

Answer: b

6. In a multiple OLS regression, what is the consequence of including an irrelevant variable?
a) The OLS estimators are biased.
b) The R-square of the regression may decrease.
c) The t statistic the OLS estimators may increase
d) The estimated variance of the OLS estimators may increase.

Answer: d

7. Suppose that the value of R^2 for an estimated regression model is exactly one. Which of the following are true?
(i) All of the data points must lie exactly on the line
(ii) All of the residuals must be zero
(iii) All of the variability of *y* have been explained by the model
(iv) The fitted line will be horizontal with respect to all of the explanatory variables

a) (ii) and (iv) only
b) (i) and (iii) only
c) (i), (ii), and (iii) only
d) (i), (ii), (iii), and (iv)

Answer: c

8. Which of the following is a nonlinear regression model?
a) $y = \beta_0 + \beta_1 x^{1/2} + u$
b) $\log(y) = \beta_0 + \beta_1 \log(x) + u$
c) $y = 1/(\beta_0 + \beta_1 x) + u$
d) $y = \beta_0 + \beta_1 x + u$

Answer: c

9. Consider the following regression model on 64 observations:

$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + u_i$

Which of the following hypotheses could we test using an *F*-test?

(i)	$H_0: \beta_2 = 0$	$H_1: \beta_2 < 0$
(ii)	$H_0: \beta_2 = 1 \; and \; \beta_3 + \beta_4 = 1$	$H_1: H_0 \; is \; not \; true$
(iii)	$H_0: \beta_3 \beta_4 = 1$	$H_1: \beta_3 \beta_4 \neq 1$
(iv)	$H_0: \beta_2 - \beta_3 - \beta_4 = 1$	$H_1: \beta_2 - \beta_3 - \beta_4 \neq 1$

a) (ii) only
b) (ii) and (iv) only
c) (i), (ii), (iii) and (iv)
d) (i), (ii), and (iv) only

Answer: b

10. You have just read a paper which claims that there is strong evidence that x affects y negatively. The researcher found that the estimated coefficient for x is significantly positive at the 5% level. You suspect that there is omitted variable bias. Which of the following statements is correct?

a) If there is omitted variable bias, and the bias is negative, then the researcher's conclusions are invalid.
b) If there is omitted variable bias, and the bias is negative, then the researcher's conclusions are still valid.
c) If there is omitted variable bias, and the bias is zero, then the researcher's conclusions are invalid.
d) If there is omitted variable bias, and the bias is positive, then the researcher's conclusions are still valid.

Answer: a

11. If the estimates of the coefficients of interest change substantially across specifications,

a) then this can be expected from sample variation.
b) then you should change the scale of the variables to make the changes appear to be smaller.
c) then this often provides evidence that the original specification had omitted variable bias.
d) then choose the specification for which your coefficient of interest is most significant.

Answer: c

12. The construction of the t-statistic for one- and a two-sided hypothesis

a) depends on the critical value from the appropriate distribution.
b) is the same.
c) is different since the critical value must be 1.645 for the one-sided hypothesis, but 1.96 for the two-sides hypothesis (5% significance level).
d) use ± 1.96 for the two-sided test, but only $+1.96$ for the one-sided test (5% significance level).

Answer: b

13. In a regression model where $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$, to test the hypothesis that $\beta_1 - 2\beta_2 = 0$ we should:

a) Regress y on $x_1, (x_1 + 2x_2), x_3$ and examine the significance of the coefficient for x_1
b) Regress y on $x_1, (x_1 + 2x_2), x_3$ and examine the significance of the coefficient for $(x_1 + 2x_2)$
c) Regress y on $x_1, (2x_1 + x_2), x_3$ and examine the significance of the coefficient for x_1
d) Regress y on $x_1, (2x_1 + x_2), x_3$ and examine the significance of the coefficient for $(2x_1 + x_2)$

Answer: c

14. Consider regression through the origin, $y = \beta_1 x + u$, which of the following statements is wrong?

a) The sum of residuals equals to 0.
b) The residual is uncorrelated with the independent variable.
c) If the true intercept parameter doesn't equal to 0, all slope estimators are biased.
d) The degree of freedom for estimating the variance of error term is $n - 1$.

Answer: a

15. In a multiple OLS regression, if you divide the explained variable y by 12 (to change units of measurement from inches to feet) which of the following will happen?

a) The estimated intercept parameter will be reduced by 12 times.
b) the estimated slope parameter will be reduced by 12 units.
c) the total sum of squares for the regression will be reduced by $\sqrt{12}$ times
d) R-squared for the regression will be reduced by 12 times.

Answer: a

16. The F-statistic for overall significance of a regression tests the null hypothesis that

a) all slope coefficients are zero.
b) all slope coefficients and the intercept are zero.
c) the intercept in the regression and at least one, but not all, of the slop coefficents is zero.
d) the slope coefficient of the variable of interest is zero, but that the other slope coefficients are not.

Answer: a

You have been commissioned to construct a model that explains NBA players' wages. The dependent variable is $lwage$, the natural log of players' annual wages (*in thousands of dollors*). Please use the following Stata regression results for the next **SIX** questions.

Source	SS	df	MS
Model	90.2698185	5	18.0539637
Residual	117.918945	263	.448361006
Total	208.188763	268	.776823743

Number of obs	= 269
F(5, 263)	= 40.27
Prob > F	= 0.0000
R-squared	= 0.4336
Adj R-squared	= 0.4228
Root MSE	= .6696

lwage	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
games	.0004132	.002682	0.15	0.878	-.0048679	.0056942
avgmin	.0302278	.0130868	2.31	0.022	.0044597	.055996
points	.0363734	.0150945	2.41	0.017	.0066519	.0660949
rebounds	.0406795	.0229455	1.77	0.077	-.0045007	.0858597
turnovers	-.0366593		-2.65	0.008	-.0637731	
_cons	5.648996	.1559075	36.23	0.000	5.34201	5.955982

17. According to the regression results above, what is the F statistic that simultaneously tests the restrictions that all slope parameters equal to 0.

a) 0.4336
b) 40.27

c) 269
d) 0.0000

Answer: b

18. According to the regression results above, what is the standard error of regression?
a) 0.6696
b) 0.448361006
c) 117.918945
d) 18.0539637

Answer: a

19. In the above regression, which of the variables are significant at the 10% level?
a) $games, avgmin, points, rebounds, turnovers$
b) $avgmin, points, rebounds, turnovers$
c) $avgmin, points, turnovers$
d) $points$

Answer: b

20. In the above regression, which of the variables show a significant **positive** effect at the 1% level?
a) $points, turnovers$
b) $points$
c) $games$
d) None of the variables are positively significant at the 1% level

Answer: b

21. According to the above regression results, what is the expected change in wages when $points$ changes, all else remaining constant?
a) Wages will increase by $36.37 when $points$ increases by 1 unit.
b) Wages will increase by 0.03637% when $points$ increases by 1%.
c) Wages will increase by $0.03637 when $points$ increases by 1 unit.
d) Wages will increase by 3.637% when $points$ increases by 1 unit.

Answer: d

22. In the above regression, what is the upper bound of 95% confidence interval of the slope parameter on $turnovers$?
a) -0.0095455
b) 0.260384
c) -0.0271138
d) -0.0637731

Answer: a

Question 2 (20 marks)

Discuss the effect of the following phenomena on the unbiasedness and the variance of the OLS estimators, $\hat{\beta}_j$:

1. Including an irrelevant variable

Effect on the unbiasedness of $\hat{\beta}_j$: Including an irrelevant variable does not affect the unbiasedness of the OLS estimators, $\hat{\beta}_j$. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: Including an irrelevant variable increases the variances of the OLS estimators of the relevant independent variables. In particular, R_j^2 increases and thus, the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, increases. **(2 marks)**

2. Omitting a relevant variable

Effect on the unbiasedness of $\hat{\beta}_j$: If the omitted relevant variable is correlated with some of the other independent variables in the model, omitting the relevant variable biases the OLS estimators, $\hat{\beta}_j$. In particular, the omitted relevant variable will be included in the error term u, and will violate the zero conditional mean assumption (Assumption 4) **(1 marks)**. If the omitted relevant variable is not correlated with any of the other independent variables in the model, omitting the relevant variable will not bias the OLS estimators, $\hat{\beta}_j$. **(1 marks)**

Effect on the variance of $\hat{\beta}_j$: Omitting a relevant variable decreases the variances of the OLS estimators of the remaining independent variables. In particular, R_j^2 decreases and thus, the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, decreases. (**Or:** We assume that variances of error term, σ^2, doesn't change when omitting a relevant variable. In fact, σ^2 increase when omitting a relevant variable, so both the denominator and numerator of variance of $\hat{\beta}_j$ increase and therefore, the variance of $\hat{\beta}_j$ can increase, decrease, or remain unchanged) **(2 marks)**

3. Perfect collinearity

Perfect collinearity means that some independent variable, x_j, is a perfect linear combination of other independent variables.

Effect on the unbiasedness of $\hat{\beta}_j$: Perfect collinearity violates Assumption 3 for the unbiasedness of OLS estimators, $\hat{\beta}_j$. In fact, under perfect collinearity, the OLS estimators, $\hat{\beta}_j$, practically cannot be computed. That's why we have to drop one of the variables in the perfect linear combination. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: If the independant variable,x_j, is perfect correlated with other independent variables, then $R_j^2 = 1$ and the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, is not defined. **(2 marks)**

4. Multicollinearity

Multicollinearity means that some independent variables are highly correlated with each other.

Effect on the unbiasedness of $\hat{\beta}_j$: Multicollinearity does not affect the unbiasedness of the OLS estimators, $\hat{\beta}_j$. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: If the independant variable,x_j, is highly correlated with other independent variables, then R_j^2 is "close to 1". Thus, the variance of $\hat{\beta}_j$, given by $var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, is large and this decreases the precision of $\hat{\beta}_j$. **(2 marks)**

5. Heteroskedasticity

Heteroskedasticity means that the error variance is not a constant and so Assumption 5 of the Gauss-Markov assumptions is violated, that is $var(u|x) \neq \sigma^2$

Effect on the unbiasedness of $\hat{\beta}_j$: Heteroskedasticity does not affect the unbiasedness of the OLS estimators, $\hat{\beta}_j$. **(2 marks)**

Effect on the variance of $\hat{\beta}_j$: Under heteroskedasticity, the usual formula for the variance of OLS estimators, $(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$, is no longer valid. **(2 marks)**

Question 3 (16 marks)

Suppose that you are interested in estimating the ceteris paribus relationship between y and x_1. For this purpose, you can collect data on two control variables, x_2 and x_3. Let $\tilde{\beta}_1$ be the regression estimate from model (1) $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$; and let $\hat{\beta}_1$ be the regression estimate from model (2) $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$.

(4 marks)

(a) Briefly explain how to use partialling out interpretation of multiple regression to find the OLS estimator $\tilde{\beta}_1$ from model (1).

The partialling out interpretation involves two steps:

1) Do a sample regression of x_1 on x_2 to obtain the residual $\tilde{r}_1$ where the effects of x_2 have been partialled out. **(2 marks)**

2) Regress y on residual $\tilde{r}_1$, and the slope coefficient represents ceteris paribus effect of x_1 on y. It can be expressed as:

$$\tilde{\beta}_1 = \frac{\sum(\tilde{r}_{i1} y_i)}{\sum \tilde{r}_{i1}^2}$$

(2 marks)

(12 marks)

(b) Prove that $\tilde{\beta}_1$ is a biased estimator if model (2) is the true population model. How large is the bias? Be specific.

$$\tilde{\beta}_1 = \frac{\sum(\tilde{r}_{i1} y_i)}{\sum \tilde{r}_{i1}^{\,2}}$$ Replace y_i with $\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + u_i$

$$= \frac{\sum(\tilde{r}_{i1}(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + u_i))}{\sum \tilde{r}_{i1}^2}$$

$$= \frac{\beta_0 \sum \tilde{r}_{i1}}{\sum \tilde{r}_{i1}^2} + \frac{\beta_1 \sum \tilde{r}_{i1} x_{i1}}{\sum \tilde{r}_{i1}^2} + \frac{\beta_2 \sum \tilde{r}_{i1} x_{i2}}{\sum \tilde{r}_{i1}^2} + \frac{\beta_3 \sum \tilde{r}_{i1} x_{i3}}{\sum \tilde{r}_{i1}^2} + \frac{\sum \tilde{r}_{i1} u_i}{\sum \tilde{r}_{i1}^2}$$ Since $\sum \tilde{r}_{i1} = 0$, $\sum \tilde{r}_{i1} x_{i2} = 0$, and $\sum \tilde{r}_{i1} x_{i1} = \sum \tilde{r}_{i1}^2$

$$= \beta_1 + \frac{\beta_3 \sum \tilde{r}_{i1} x_{i3}}{\sum \tilde{r}_{i1}^2} + \frac{\sum(\tilde{r}_{i1} u_i)}{\sum \tilde{r}_{i1}^2}$$

(6 marks)

Take expectation on $\tilde{\beta}_1$, conditioning on X, we have

$$E(\tilde{\beta}_1) = \beta_1 + E\left[\frac{\beta_3 \sum \tilde{r}_{i1} x_{i3}}{\sum \tilde{r}_{i1}^2}\right] + E\left[\frac{\sum(\tilde{r}_{i1} u_i)}{\sum \tilde{r}_{i1}^2}\right]$$

$$= \beta_1 + \frac{\beta_3 \sum \tilde{r}_{i1} x_{i3}}{\sum \tilde{r}_{i1}^2} + \frac{\sum(\tilde{r}_{i1} E(u_i))}{\sum \tilde{r}_{i1}^2}$$ Since $E(u_i|X) = 0$

$$= \beta_1 + \frac{\beta_3 \sum \tilde{r}_{i1} x_{i3}}{\sum \tilde{r}_{i1}^2}$$

(4 marks)

Therefore, $\tilde{\beta}_1$is a biased estimator of the slope coefficient β_1, and the omitted variable bias is $\frac{\beta_3 \sum \tilde{r}_{i1} x_{i3}}{\sum \tilde{r}_{i1}^2}$

(2 marks)

Question 4 (20 points)

You have been commissioned to investigate the relationship between the annual salaries of Chief Executive Officers (CEOs) of firms and the annual profits of their firms. The dependent variable is $salary_i$, the annual salary of the CEO of the i-th firm, measured in *thousands of dollars* per year. The explanatory variable is $profits_i$, the annual profits of the i-th firm, measured in *millions of dollars per year*. The model you propose to estimate is given by the population regression equation

$$salary_i = \beta_0 + \beta_1 profits_i + u_i$$

Your research assistant has used observations on $salary_i$ and $profits_i$ for a sample of 177 corporations to estimate the following OLS sample regression equation, where the figures in parentheses below the coefficient estimates are the *standard errors* of the coefficient estimates:

$$\underset{\quad\quad\quad

$\widehat{\beta_1} - c_\alpha se(\widehat{\beta_1}) \leq \beta_1 \leq \widehat{\beta_1} + c_\alpha se(\widehat{\beta_1})$ **(2 marks)**

Degree of freedom $= n - 2 = 177 - 2 = 175$
Critical value of t_{175} at 5% significance level ($\alpha = 0.01$) is $\boldsymbol{c_{0.05} = 1.960}$

$c_\alpha se(\widehat{\beta_1}) = 1.960 * 0.10094 = 0.1978$

Lower 95% confidence limit for β_1 is: **(2 marks)**

$\widehat{\beta_1} - c_\alpha se(\widehat{\beta_1}) = 0.5723 - 0.1978 = \mathbf{0.3745}$

Upper 95% confidence limit for β_1 is: **(2 marks)**

$\widehat{\beta_1} + c_\alpha se(\widehat{\beta_1}) = 0.5723 + 0.1978 = \mathbf{0.7701}$

Result: The two-sided 95% confidence interval for β_1 is: [0.3745, 0.7701]

(7 points)
(c) Perform a test of the proposition that an increase in profits of 1 million dollars per year is associated on average with an increase in CEO salary of 500 dollars per year. Use the 5 percent significance level (i.e., $\alpha = 0.05$). State the null hypothesis H_0 and the alternative hypothesis H_1. Show how you calculated the test statistic. State the decision rule you use, and the inference you would draw from the test.

Null hypothesis: $H_0: \beta_1 = 0.5$
Alternative hypothesis: $H_1: \beta_1 \neq 0.5$ a ***two-tail*** **test** **(1 marks)**

t-test statistic under H_0 is: **(3 marks)**

$$t(\widehat{\beta_1}) = \frac{\widehat{\beta_1} - \beta_1}{se(\widehat{\beta_1})} = \frac{\widehat{\beta_1} - 0.5}{se(\widehat{\beta_1})} = \frac{0.57230 - 0.5}{0.10094} = \mathbf{0.716}$$

Decision Rule: This is a ***two-tail*** **test**. Compare the sample t statistic with the α-level critical value of the t_{175} distribution. **(1 marks)**
1. If $|t| \geq c_\alpha$, do not reject H_0 at the significance level α.
2. If $|t| < c_\alpha$, reject H_0 at the significance level α.

Degree of freedom $= n - 2 = 177 - 2 = 175$
Critical value of t_{175} at 1% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 1.960}$ **(1 marks)**

Inference: **(1 marks)**
Since $\boldsymbol{t = 0.716 < 1.960 = c_{0.05}}$**, do not reject $\mathbf{H_0}$** at the **5%** significance level.

Question 5 (30 marks)

You are investigating the relationship between the final exam grades of university students in an introductory economics course and those students' class attendance, as measured by the percentage of classes each student attended during the term. You also have sample data on two additional explanatory variables: each student's cumulative GPA (Grade Point Average) prior to the term in which the introductory economics course was taken; and each student's score on a standardized college entrance exam, the ACT exam. You have sample data for 680 students on the following variables:

$finalpct_i =$ final exam grade of the i-th student, measured in *percentage points*;
$attrate_i =$ percentage of classes attended by the i-th student during the term, measured in *percentage points*;
$GPA_i =$ Cumulative Grade Point Average (GPA) of the i-th student prior to the term in which the introductory economics course was taken, measured *out of 4.0*;
$ACT_i =$ ACT score of the i-th student on the ACT college entrance exam, measured in *points*.

Using the given sample data on 680 students, your trusty research assistant has estimated regression equation (1) and obtained the following estimation results (with estimated standard errors given in parentheses below the coefficient estimates):

$$finalpct_i = \beta_0 + \beta_1 attrate_i + \beta_2 GPA_i + \beta_3 GPA_i^2 + \beta_4 ACT_i + u_i \qquad (1)$$

$\hat{\beta}_0 = 55.564$ (8.077) $\quad \hat{\beta}_1 = 0.071698$ (0.0279) $\quad \hat{\beta}_2 = -18.6518$ (5.5788)

$\hat{\beta}_3 = 4.4785$ (1.0513) $\quad \hat{\beta}_4 = 0.9$ (0.1336)

$\hat{Cov}(\hat{\beta}_1, \hat{\beta}_2) = -0.024 \quad \hat{Cov}(\hat{\beta}_1, \hat{\beta}_3) = 0.00203 \quad \hat{Cov}(\hat{\beta}_1, \hat{\beta}_4) = 0.001284$

$\hat{Cov}(\hat{\beta}_2, \hat{\beta}_3) = -5.7846 \quad \hat{Cov}(\hat{\beta}_2, \hat{\beta}_4) = 0.075667 \quad \hat{Cov}(\hat{\beta}_3, \hat{\beta}_4) = -0.025335$

$SSR = 73218.11 \quad SST = 94137.17 \quad N = 680$

SSR is the Residual Sum-of-Squares and SST is the Total Sum-of-Squares for sample regression equation (1). Sample size $N = 680$. $\hat{Cov}(\hat{\beta}_j, \hat{\beta}_h)$ is the estimated covariance between coefficient estimates $\hat{\beta}_j$ and $\hat{\beta}_h$. Estimated standard errors are given in parentheses below the coefficient estimates $\hat{\beta}_j$ $(j = 0,1, \dots, 4)$

(10 marks)

(a) Compute a test of the proposition that students' cumulative Grade Point Average has no effect on students' final exam grade. State the coefficient restrictions on regression equation (1) implied by this proposition; that is, state the null hypothesis H_0 and the alternative hypothesis H_1. Write the restricted regression equation implied by the null hypothesis H_0. OLS estimation of this restricted regression equation yields a Residual Sum-of-Squares value of $SSR = 78125.22$. Use this information, together with the results from OLS estimation of equation (1), to calculate the required test statistic. State the decision rule you use, and the inference you would draw from the test at the 5 percent significance level.

Test: $H_0: \beta_2 = 0, \beta_3 = 0$ versus: H_1: H_0 is not true **(2 marks)**

Restricted regression implied by H_0: **(1 marks)**

$$finalpct_i = \beta_0 + \beta_1 attrate_i + \beta_4 ACT + u_i$$

Compute F statistic:

(4 marks)

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)} = \frac{(78125.22 - 73218.11)/2}{73218.11/(680-4-1)} = \mathbf{22.62}$$

Decision Rule: Compare the sample F statistic with the α-level critical value of the $\mathbf{F_{2,675}}$ distribution. **(1 marks)**

1. If $F \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $F > c_\alpha$, reject H_0 at the significance level α.

Critical value of $\mathbf{F_{2,675}}$ at 5% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 3.00}$ **(1 marks)**

Inference: **(1 marks)**

Since $\boldsymbol{F = 22.62 > 3 = c_{0.05}}$**, reject $\mathbf{H_0}$** at **5%** significance level.

(11 marks)

(b) The average student in the course has a cumulative Grade Point Average of 2.6; that is, the sample mean value of GPA_i equals 2.6. Write the expression (or formula) for the marginal effect on $finalpct_i$ of GPA_i implied by regression equation (1). Use the estimation results for regression equation (1) to test the proposition that the marginal effect of cumulative GPA (GPA_i) on students' final exam grade ($finalpct_i$) equals zero for the average student whose GPA is 2.6. State the null and alternative hypotheses, and show how you calculate the required test statistic. State the inference you would draw from the test at the 5 percent significance level.

Proposition to test: The marginal effect of GPA_i with respect to $finalpct_i$ is equal to zero for $GPA_i = 2.6$

The **marginal effect of $\boldsymbol{GPA_i}$** is: **(2 marks)**

$$\frac{\partial finalpct_i}{\partial GPA_i} = \beta_2 + 2\beta_3 GPA_i$$

Test: $H_0: \beta_2 + 2 * 2.6\beta_3 = \beta_2 + 5.2\beta_3 = 0$ versus:
$H_1: \beta_2 + 5.2\beta_3 \neq 0$ ***two-tails t-test*** **(2 marks)**

Compute t-statistic: **(4 marks)**

$\hat{\beta}_2 + 5.2\hat{\beta}_3 = -18.6518 + 5.2 * 4.4785 = \mathbf{4.6364}$

$$\hat{Var}(\hat{\beta}_2 + 5.2\hat{\beta}_3) = \hat{Var}(\hat{\beta}_2) + (5.2)^2 \hat{Var}(\hat{\beta}_3) + 10.4 cov(\hat{\beta}_2, \hat{\beta}_3)$$
$$\hat{Var}(\hat{\beta}_2 + 5.2\hat{\beta}_3) = (5.5788)^2 + (27.04)(1.0512)^2 + 10.4(-5.7846)$$
$$\hat{Var}(\hat{\beta}_2 + 5.2\hat{\beta}_3) = \mathbf{0.843}$$

$$se\left(\hat{\beta}_2 + 5.2\hat{\beta}_3\right) = \sqrt{\hat{Var}\left(\hat{\beta}_2 + 5.2\hat{\beta}_3\right)} = \mathbf{0.918}$$

$$t = \frac{\hat{\beta}_2 + 5.2\hat{\beta}_3}{se\left(\hat{\beta}_2 + 5.2\hat{\beta}_3\right)} = \frac{4.6364}{0.918} = \mathbf{5.051}$$

Decision Rule: This is a ***two-tail*** **test**. Compare the sample t statistic with the α-level critical value of the t_{272} distribution. **(1 mark)**

1. If $|t| \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $|t| > c_\alpha$, reject H_0 at the significance level α.

Critical value of t_{675} at 5% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 1.96}$ **(1 mark)**

Inference: **(1 mark)**

Since $\boldsymbol{t = 5.051 > 1.96 = c_{0.05}}$**, reject $\mathbf{H_0}$** at the **5%** significance level.

(9 marks)

(c) Use the estimation results for regression equation (1) to test the joint significance of the slope coefficient estimates $\hat{\beta}_j$ $(j = 1, \ldots, 4)$ in regression equation (1). Perform the test at the 5 percent significance level (i.e., for significance level $\alpha = 0.05$). State the null and alternative hypotheses, and show how you calculate the required test statistic. State the inference you would draw from the test at the 5 percent significance level.

This is an overall significance test:

Test: $H_0: \beta_1 = 0, \beta_2 = 0, \beta_3 = 0, \beta_4 = 0$ versus H_1: H_0 is not true **(2 marks)**

Compute F statistic: **(4 marks)**

$$R^2_{ur} = 1 - \frac{SSR}{SST} = 1 - \frac{73218.11}{94137.17} = \mathbf{0.2222}$$

$$F = \frac{(R^2_{ur} - R^2_r)/q}{(1 - R^2_{ur})/(n - k - 1)} = \frac{(0.2222 - 0)/4}{(1 - 0.2222)/(680 - 4 - 1)} = \mathbf{48.21}$$

Decision Rule: Compare the sample F statistic with the α-level critical value of the $\mathbf{F_{4,675}}$ distribution. **(1 marks)**

1. If $F \leq c_\alpha$, do not reject H_0 at the significance level α.
2. If $F > c_\alpha$, reject H_0 at the significance level α.

Critical value of $\mathbf{F_{4,675}}$ at 5% significance level ($\alpha = 0.05$) is $\boldsymbol{c_{0.05} = 2.37}$ **(1 marks)**

Inference: **(1 marks)**

Since $\boldsymbol{F = 48.21 > 2.37 = c_{0.05}}$**, reject $\mathbf{H_0}$** at **5%** significance level.